RESEARCH PROPOSAL PRESENTATION

PREDICTING HEART DISEASE WITH MACHINE LEARNING: PERFORMANCE, EXPLAINABILITY & CLINICAL TRUST

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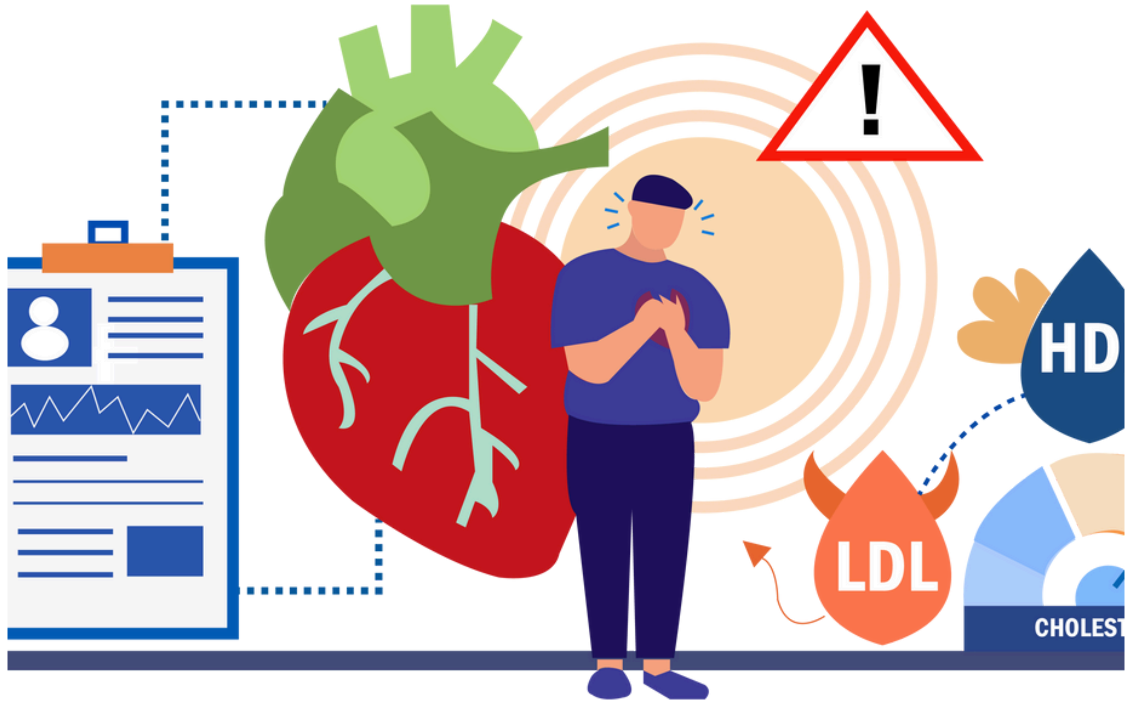


HELLO!

This presentation explores the use of machine learning techniques in predicting heart disease risk, highlighting methodologies, findings, and their impact on clinical practices and patient outcomes.

TABLE OF CONTENT

BACKGROUND OF THE STUDY

HEART DISEASE IMPACT

Heart disease remains a leading cause of death globally, affecting millions and straining healthcare systems.Heart disease is the leading global killer (≈31% of all deaths)
Early detection drastically improves survival

MACHINE LEARNING POTENTIAL

Machine learning offers innovative approaches to improve early detection and prediction of heart disease risks in patients. ML can spot complex patterns in routine patient data that clinicians might miss

RESEARCH NECESSITY

Understanding the relationship between machine learning and heart disease is crucial for advancing medical research and patient care.

LITRATURE REVIEW

HUSAM & AL-QURAAN (2022)

• **Tested** DT, LR, NB, KNN & SVM on the Cleveland dataset
• **Best:** SVM at 87.5% accuracy
• **Limitation:** small sample, no feature-engineering

SHAHEEN ET AL. (2024)

• **Used** RF, LR & DT on Jordan University Hospital records
• **Best:** RF at 93.6% accuracy after careful preprocessing
• **Limitation:** single-site data, unknown generalisability

AL-ALSHAIKH ET AL. (2024)

• Hybrid deep + traditional ML (ML-HDPM)
• 95.5% train / 89.1% test accuracy
• **Limitation:** model complexity may hinder clinical use

LITRATURE REVIEW

EL-SOFANY ET AL. (2024)

- XGBoost + SHAP for global explanations
- 97.6% **accuracy** with feature-attribution
- **Limitation:** heavy compute, no real-time validation

BOUHAMED ET AL. (2024)

- Ensemble risk-scoring tool for early intervention
- High overall **accuracy**, built for real-world use
- **Limitation:** scarce detail on individual model performance

RESEARCH GAP

- **Performance vs. Explainability**

Prior studies focus on accuracy but seldom integrate clear, clinician-friendly explanations.

- **Single-Cohort Validation**

Most models are trained and tested on one dataset, limiting confidence in new patient populations.

DATASETS FEAUTURE AND PREPROCESSING

Feature	Description
Age	how old the patient is.
Sex	if the patient is male or female.
ChestPainType	what kind of chest pain the person feel.
RestingBP	blood pressure when the person is resting.
Cholesterol	level of cholesterol in blood.
FastingBS	if blood sugar more than 120 (1 = yes, 0 = no).
RestingECG	result of ECG test when person is resting.
MaxHR	the max heart rate when person do exercise.
ExerciseAngina	if person feel angina while exercise (Y = yes, N = no).
Oldpeak	a value from stress test show how much heart slow down.
ST_Slope	the shape of ST segment from ECG (Up, Flat, Down).

- **Dataset:** 303 patients, 11 features + HeartDisease label
- **Missingor Zero Handling:** replaced 0's in RestingBP & Cholesterol with median
- **Encoding:** label-encoded Sex, ChestPainType, RestingECG, ExerciseAngina, ST_Slope
- **Split:** 80% train / 20% test, random_state=42 for reproducibility

RESEARCH QUESTION AND OBJECTIVE

- Which ML model best predicts heart disease from patient records, and how can we make its predictions transparent and trustworthy to clinicians?
- Drives both our quantitative experiments and our expert interviews

01

SCOPE OF THE STUDY

My research proposal attempts to identify problems and opportunities, conduct feasibility assessment, and provide strategic intelligence.

02

RELEVANCE OF THE STUDY

The businesses that want to remain competitive and address the changing demands of customers should understand its market implications.

03

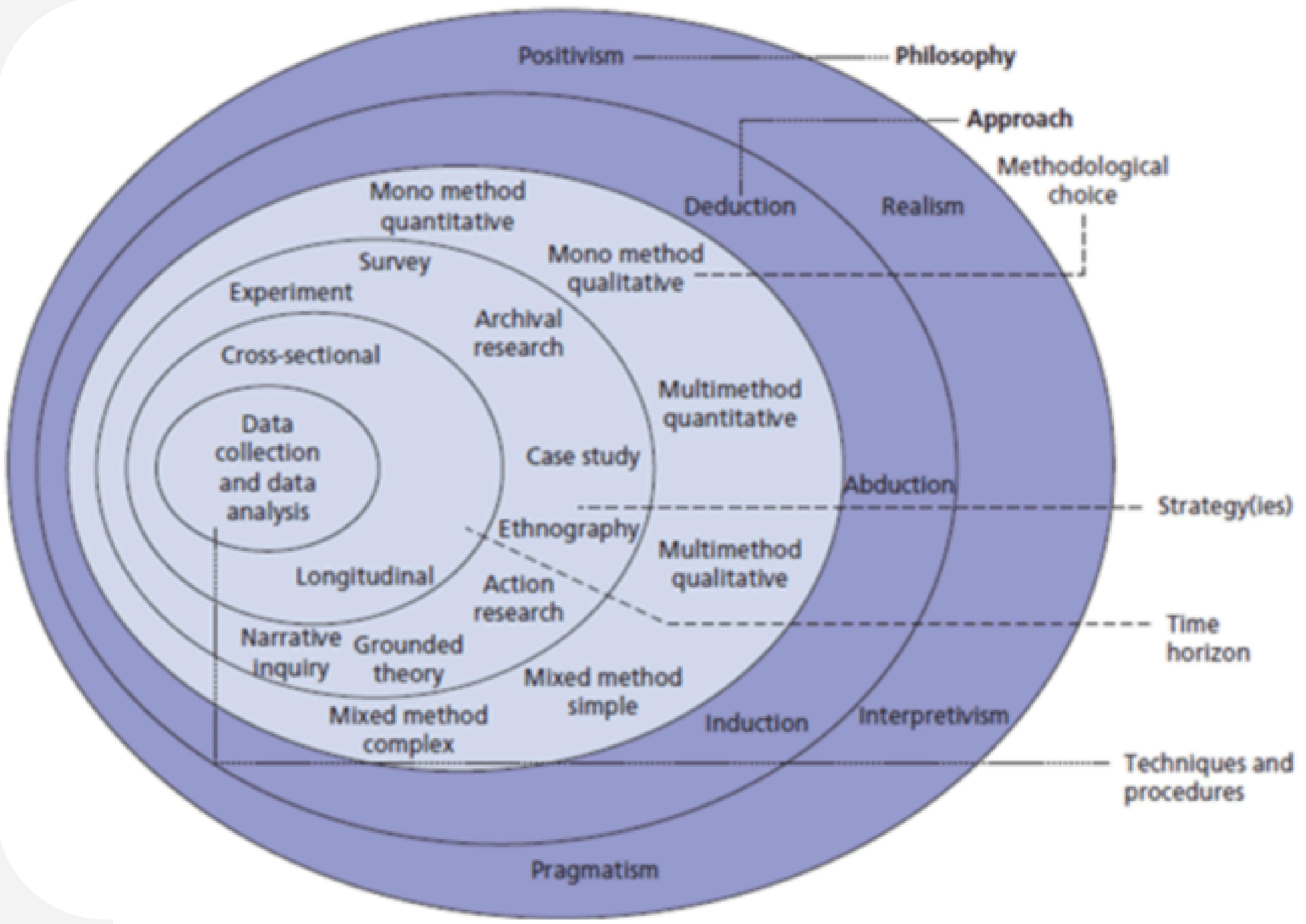
OBJECTIVE

- Comparing three classifiers (Logistic Regression, Decision Tree, Random Forest) on the heart-disease dataset for predictive performance.
- Applying SHAP to explain the best model's decisions and build clinician trust.

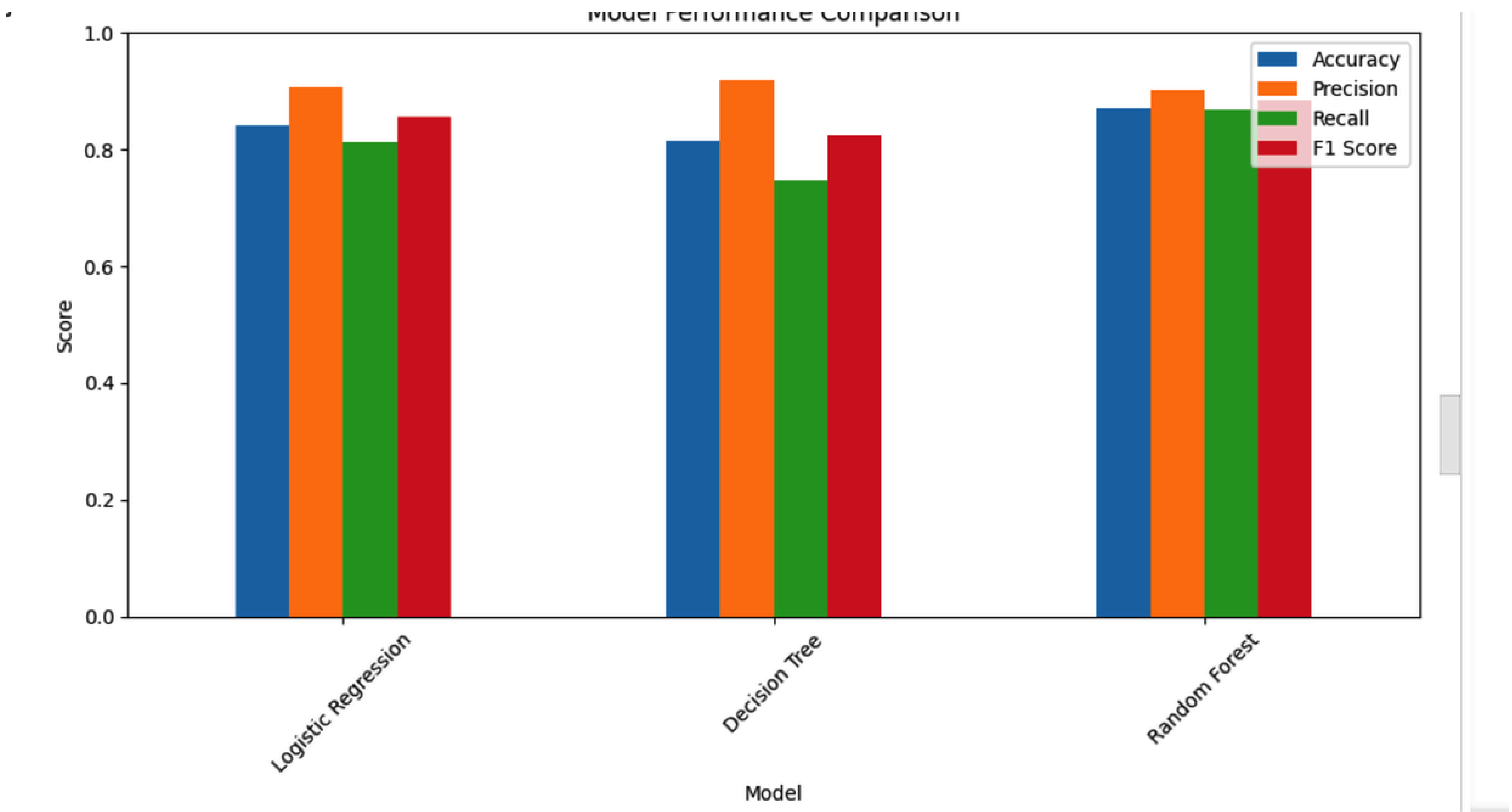
METHODOLOGY

QUALITATIVE METHODS

- Mixed-methods design:
- Quantitative strand: train & test three classifiers on secondary “heart.csv” data
- Qualitative strand: semi-structured interviews with a cardiologist & an AI expert
- Philosophy: Pragmatism → lets us blend numeric accuracy checks with clinician feedback



SLIDE 7 – KEY FINDINGS: PERFORMANCE COMPARISON



LOGISTIC REGRESSION

- Accuracy: 0.84
- Precision: 0.91
- Recall: 0.81
- F1 Score: 0.857

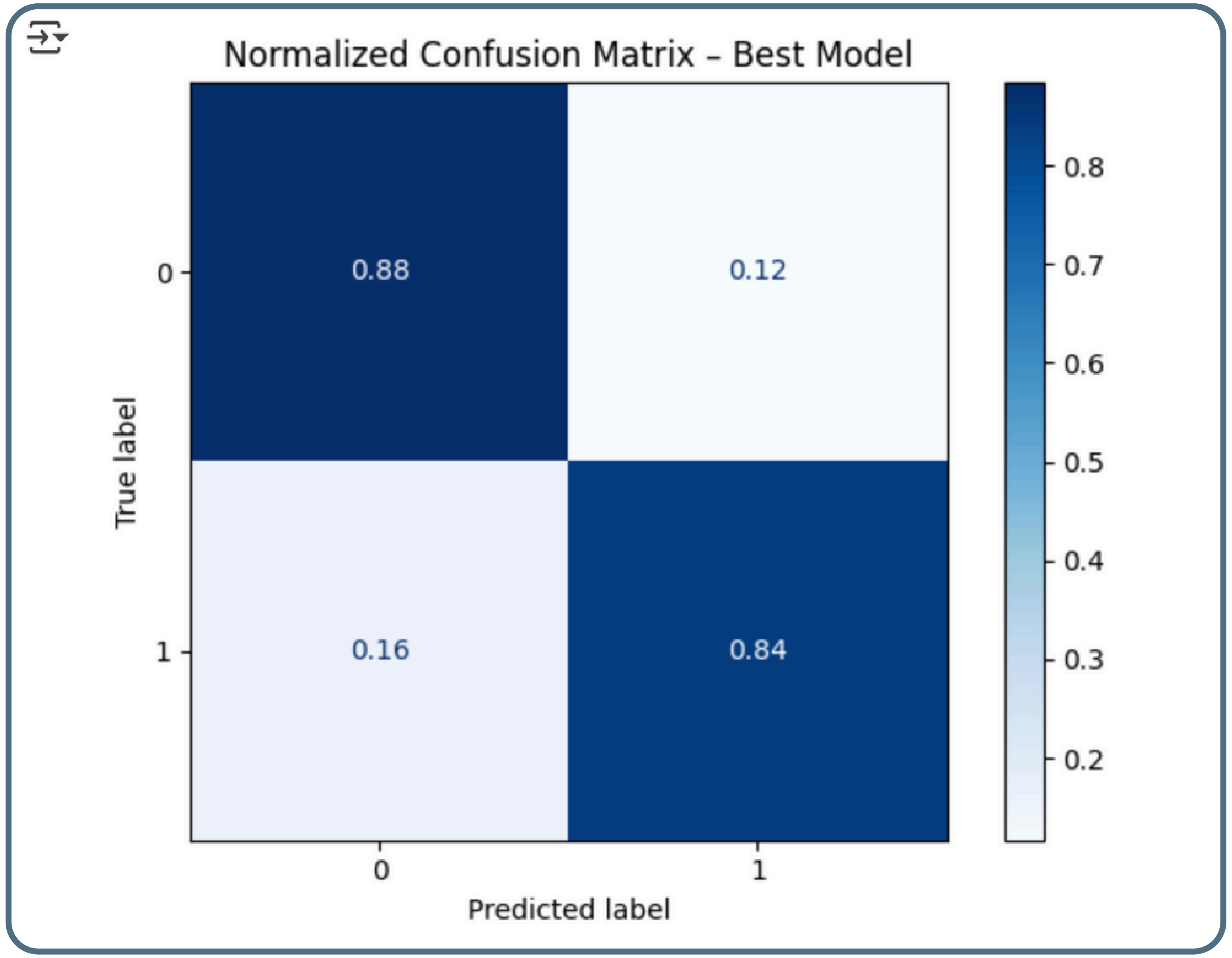
DESCISSION TREEE

- Accuracy: 0.82
- Precision: 0.92
- Recall: 0.75
- F1 Score: 0.825

RANDOM FOREST

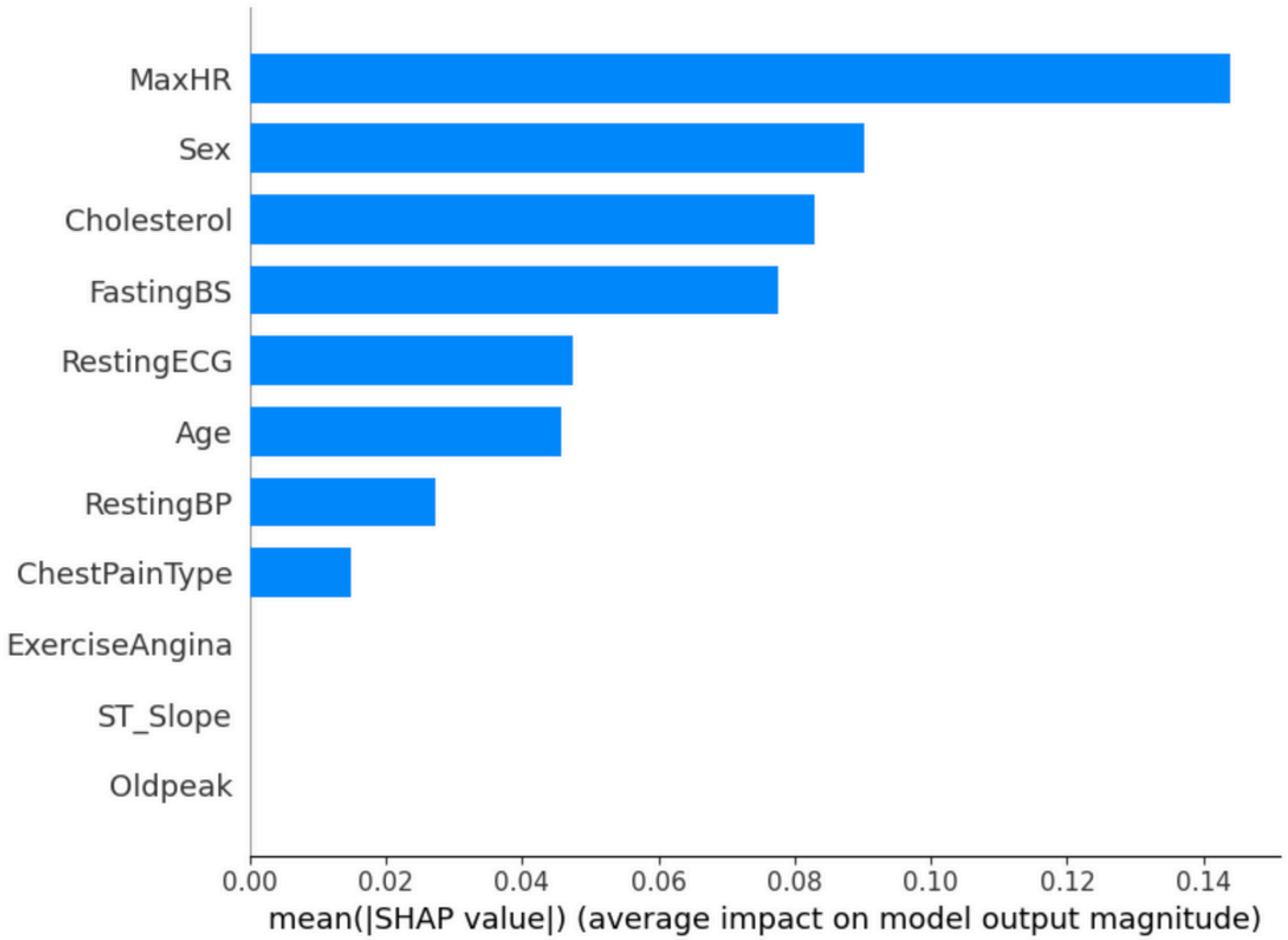
- Accuracy: 0.87
- Precision: 0.90
- Recall: 0.87
- F1 Score: 0.886

KEY FINDINGS: CONFUSION MATRIX



- True positives (disease): 84% correctly flagged
- True negatives (healthy): 88% correctly identified
- False negatives: 16% → clinical reminder to adjust sensitivity

KEY FINDINGS: EXPLAINABILITY (SHAP)



- Top **drivers** of “disease” prediction:
  a. MaxHR
  b. Sex
  c. Cholesterol

- Builds clinician trust by showing why the model decided as it did

CONCLUSION & RECOMMENDATIONS

- **Conclusion**

Tuned Random Forest proved the most accurate predictor of heart disease, and SHAP analyses confirmed that its top features (MaxHR, Sex, Cholesterol) make clinical sense.

- **Recommendations**

1. External validation on new hospital datasets to ensure the model generalizes.
2. Add richer clinical details (medication history, lifestyle data, imaging) to boost prediction power.
3. Build an interactive dashboard that pairs model output with SHAP explanations so clinicians can explore & trust each prediction.

REFLECTION ON RESEARCH PROCESS

- **What I Did & Learned**
- I combined quantitative experiments (Logistic Regression, Decision Tree, Random Forest) with semi-structured interviews to balance performance testing and clinician insight.
- Cleaning the data (replacing impossible zeros with medians) taught me how small errors can derail model accuracy.
- Training and tuning the three models helped me see the trade-off between high accuracy and overfitting.
- Using SHAP revealed exactly which patient features drove each prediction, deepening my grasp of explainable AI.
- Conducting interviews pushed me to craft clear questions, take structured notes, and group findings into themes around trust and transparency.

SWOT Analysis of My Research Experience

strength

- Python & ML skills
- SHAP explainability
- Integrated expert insights

weakneses

- One dataset only
- Only 2 experts
- Time limited modeling

Opportunities

- Add clinical features
- Build a dashboard
- Validate on new data

threats

- Confirmation bias
- Overfitting risks
- Limited generalizability

Key Insights and Reflections

Through this research process, I gained valuable insights into the balance between technical modeling and qualitative interviews, emphasizing the importance of a well-rounded approach to predicting heart disease using machine learning techniques.

Data Quality

Ensuring clean data is vital for accurate model performance and reliability.

Mixed-Methods Approach

The mixed-methods approach taught me to effectively balance modeling with interviews, enhancing the overall quality of my project. enhances depth and understanding of the findings.

Interdisciplinary Collaboration

Engaging with experts adds credibility, but more voices could improve reliability.

THANK YOU